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Materials Research Express

RECEIVED
20 February 2019

REVISED
24 March 2019

ACCEPTED FOR PUBLICATION
16 April 2019

PUBLISHED
3 May 2019

PAPER

Thermodynamic optimization of the Ni–Al–Pr ternary system

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Keywords: Ni–Al–Pr, CALPHAD, miedema theory, thermodynamic optimization, NiAl alloy

Abstract

A thermodynamic optimization of the Ni–Al–Pr ternary system that includes eight ternary phases was conducted. The Al–Pr binary system was re-optimized thermodynamically with a consideration for the Pr solubility in fcc(Al). The enthalpy of formation of eight ternary intermetallic compounds was calculated by using Miedema theory. A self-consistent set of thermodynamic parameters for the Al–Pr binary and Ni–Al–Pr ternary systems was optimized by using the calculation of phase diagram (CALPHAD) method. The calculated results for the binary and ternary systems showed satisfactory agreement with the experimental literature data. This work is meaningful in the design of NiAl alloys.

1. Introduction

The intermetallic compound NiAl is one of the most promising high-temperature structural materials because of its good high-temperature strength, resistance to oxidation and corrosion, relatively low material density and high melting point. However, NiAl-alloy application is limited because it is difficult to process due to brittle fractures and a low tensile ductility result at room temperature [1, 2]. Alloying is an effective method to improve the mechanical properties of NiAl alloys [3]. Alloying elements, such as, W, V, Ti, Zr, Cr, Mn and Fe have been added to the NiAl alloy and the ductility has improved. These alloy-element additions, however, negate many of the original advantages, such as a high melting point and oxidation resistance [4, 5]. Rare-earth elements (REEs) can be used as active elements to improve the oxidation resistance, and as alloying elements they are more attractive and improve the material properties significantly. Guo *et al* has been studied the effects of REEs (such as La, Y, Nd and Ce) on the microstructure and properties of NiAl and NiAl–Cr(Mo) alloys. It was found that the addition of a small amount of REEs can improve the room-temperature compressive ductility of the NiAl alloy without diminishing other essential properties [6]. REEs addition must be limited within a certain range and the appropriate amount of REEs additions lies between 0.05–0.1 wt%. When the content of the REEs addition is more than 0.1 wt%, the new phase would form and deteriorate the alloy's properties [7]. Therefore, it is important to controls the addition of REEs accurately.

In general, the design of NiAl alloys is achieved by trial and error, which means that the researcher selects the material composition and process parameters through extensive experimentation and experience [8, 9]. This approach consumes extensive manpower and material resources, and is not conducive to systematic exploration. The effects of rare-earth elements (Sc, Y, La, Ce, Nd, Pr, Sm, Eu and Pm) on the atomic level with a crystal structure, elastic properties and electronic structure of NiAl intermetallic compounds were studied by He *et al* using the first-principles method [10–12]. However, limited thermodynamic studies exist on the effect of rare-earth elements on NiAl intermetallic compounds. As a 'map' of material design, the phase diagram has an important guiding significance to develop new NiAl alloys. The CALPHAD [13] method has become an effective way for multi-alloy design and it is used extensively in the design and development of new materials [14]. The building of a thermodynamic database of NiAl alloys that contain REEs will be research work with significant theoretical value to develop high-performance NiAl alloys. Thermodynamic studies of Ni–Al–REE

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(REE = Y, La, Ce, Nd) ternary systems have been carried out in previous work [15–18]. In this study, the thermodynamic optimization of the Ni–Al–Pr ternary system was performed by using the CALPHAD method.

2. Literature review

2.1. Al–Pr binary system

Phase-equilibrium [19–22] and thermodynamic [23–26] information of the Al–Pr binary system have been investigated by many authors. Buschow *et al* [19] first determined the entire composition by metallography, thermal analysis and x-ray diffractometry, and found four solution phases (liquid, fcc, bcc and dhcp) and six intermetallic compounds (Al_4Pr, Al_3Pr, Al_2Pr, AlPr, $AlPr_2$ and $AlPr_3$), including an allotropic transformation α_Al_4Pr/β_Al_4Pr at 965 °C. Later, Buschow corrected the phase to $Al_{11}Pr_3$ by using a full-matrix least-squares refinement on the basis of a threefold unit cell of $A1_4Ba$ [20]. The slight solubility of Pr in the (Al) solution was found to be up to 0.004 at% at 600 °C and 0.008 at% at 620 °C by Drits *et al* [21]. A thermodynamic optimization of the Al–Pr system was conducted by Jin *et al* [27] and Yin *et al* [28] and a modified quasichemical model was applied to the liquid phase in the work of Jin [27], which is not applicable in the pan-nickel database. The Al–Pr phase diagram and thermodynamic evaluation that were calculated by Yin *et al* [28] agreed well with the literature data, but they did not treat the $Al_{11}Pr_3$ phase as β_$Al_{11}Pr_3$ and α_$Al_{11}Pr_3$. We added an allotropic transformation on Yin's basis and adjusted the parameters of the $Al_{11}Pr_3$ phase slightly. We also considered the solubility of Pr in the Al solid solution.

2.2. Ni–Pr binary system

The Ni–Pr binary system has been studied mainly by Huang *et al* [29] and Pan *et al* [30], and assessed by Okamoto [31]. This system exists as four solution phases (liquid, fcc, bcc and dhcp) and seven intermetallics (Ni_3Pr_7, $NiPr_3$, NiPr, Ni_2Pr, Ni_3Pr, Ni_7Pr_2 and Ni_5Pr). Recently, the Ni–Pr binary system has been optimized by Rahou *et al* [32] and the optimization results agree well with the literature data. Therefore, we have used Rahou's thermodynamic parameters.

2.3. Ni–Al binary system

The Ni–Al binary system has been assessed in our previous work [15] based on Huang's thermodynamic parameters [33]. The thermodynamic parameters of the Ni–Al system have been taken from previous work without changes.

2.4. Ni–Al–Pr ternary system

Ni–Al–Pr ternary system was assessed by Ferro *et al* [34]. The isothermal cross section of the Ni–Al–Pr ternary system was in the range of 0 at%–33 at% Pr, except for the Al-rich region at 800 °C, which was measured by Rykhal *et al* [35] using x-ray diffraction and microscopy methods. In his research, the existence of seven new intermetallic compounds was confirmed for the first time, and their composition and crystal structure were described. They reported that the Ni_5Pr binary phase exhibits an extended solid solubility for fcc (Al), up to 35 at%. The solubility of fcc (Al) in Ni_2Pr as determined by Leo *et al* [36] was 4 at%. Later, a new ternary compound, $NiAl_2Pr$, was identified by Bruzzone *et al* [37]. Crystal information on the ternary compounds of the Ni-Al-Pr ternary system is presented in table 1. Thermodynamic information on the Ni–Al–Pr ternary systems is rare. The enthalpy of formation of the ternary intermediate τ_7 (NiAlPr) phase was determined by Borzone *et al* [38] using high-temperature direct calorimetry.

Eight ternary intermediate phases existed in the isothermal section at 800 °C of the Ni–Al–Pr ternary system. Experimental data exist only for the τ_7 (NiAlPr) phase. In the absence of experimental data, Miedema's theory and DFT calculation provides a supplement to the CALPHAD method. With developments and improvements after several decades, Miedema's model has become one of the most successful theoretical models to predict the thermodynamic properties of alloys, and it can better predict the formation of binary alloys [39, 40]. Extensions for multi-component alloy systems have been proposed by many authors [41–49]. Later, the extensions of Miedema's model for multi-component alloy systems were simplified by Ouyang *et al* [50, 51]. What's more, the free first-principles databases (OQMD [52] and Materials Project [53]) used an *ab initio* method to calculate the enthalpies of formation of some ternary intermediates. We used Ouyang's model with Miedema theory to calculate the formation enthalpy of a ternary phase and the DFT calculated results provided by first-principles databases to supplement the CALPHAD method.

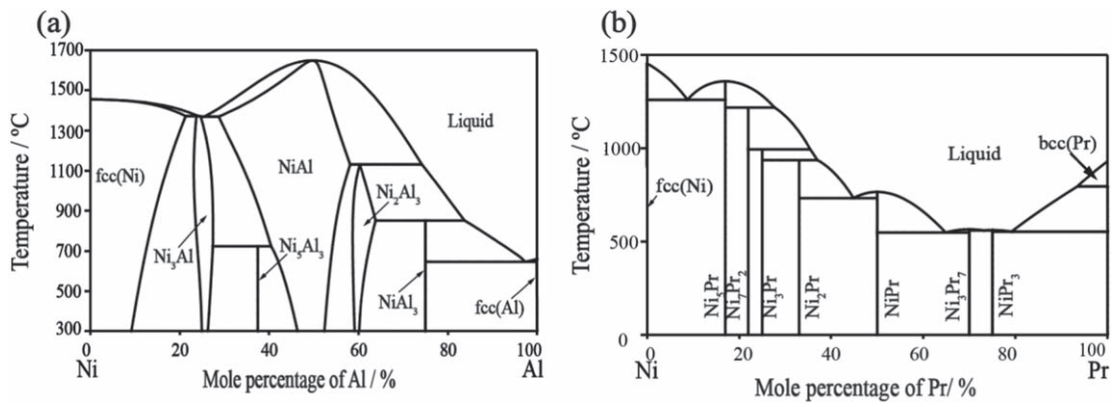

Figure 1. Calculated Ni–Al phase diagram (a) and calculated Ni–Pr phase diagram (b).

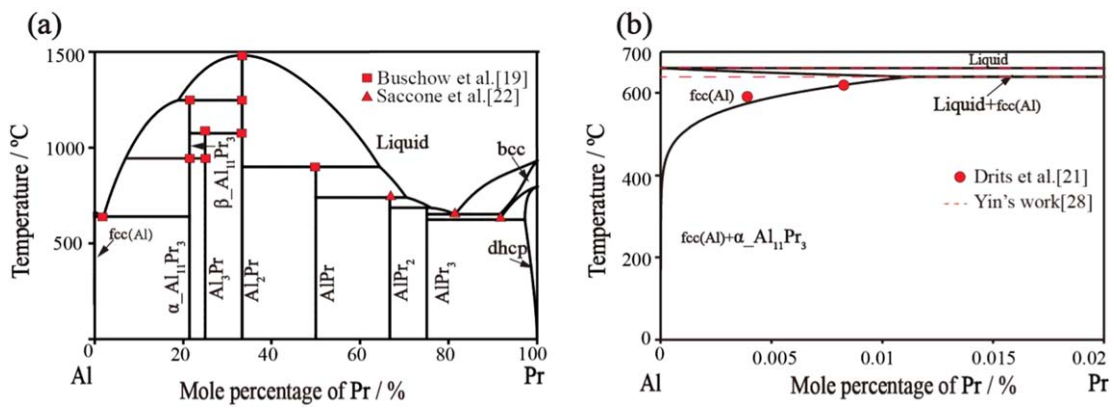

Figure 2. Calculated Al–Pr phase diagram (a) Full composition range; (b) magnified Al-rich corner showing solubility of Pr in fcc (Al).

Table 1. Ternary intermetallic compounds of Ni–Al–Pr ternary system.

Phase		Prototype	Composition (at%)	Source
τ_1	Ni_2Al_7Pr	—	70.00 Al, 10.00 Pr	[35]
τ_2	Ni_2Al_5Pr	$PrNi_2Al_5$	62.50Al, 12.50 Pr	[35]
τ_3	$NiAl_4Pr$	YNiAl4	66.66 Al, 16.67 Pr	[35]
τ_4	Ni_3Al_2Pr	$CaCu_5$	38.33 Al, 16.67 Pr	[35]
τ_5	$NiAl_2Pr$	$CuMgAl_2$	50.00 Al, 25.00 Pr	[37]
τ_6	Ni_8AlPr_3	Ce_3Co_8Si	8.33 Al, 25.00 Pr	[35]
τ_7	NiAlPr	ZrNiAl	33.33 Al, 33.33 Pr	[35]
τ_8	Ni_2AlPr_2	Mo_2NiB_2	20.00 Al, 40.00 Pr	[35]

3. Thermodynamic model

3.1. Solution phase

Pandat software [54] was used in the present work. The liquid, fcc, bcc and dhcp phases were treated with a substitutional solution model for the Gibbs free energy G expression:

$$G_m^{\varphi} = \sum_{i=Al,Ni,Pr} (x_i {}^0G_i^{\varphi}) + RT \sum_{i=Al,Ni,Pr} (x_i \ln(x_i)) + {}^{ex}G_m^{\varphi} \quad (1)$$

in which i represents elements Al, Pr and Ni; x denotes the mole fraction, R is the gas constant and T is the temperature in K; ${}^0G_i^{\varphi}$ is the molar Gibbs free energy for pure elements i in the structure φ phase from Dinsdale [55] and ${}^{ex}G_m^{\varphi}$ is the excess Gibbs free energy, which is expressed in the Redlich–Kister polynomial [56] as:

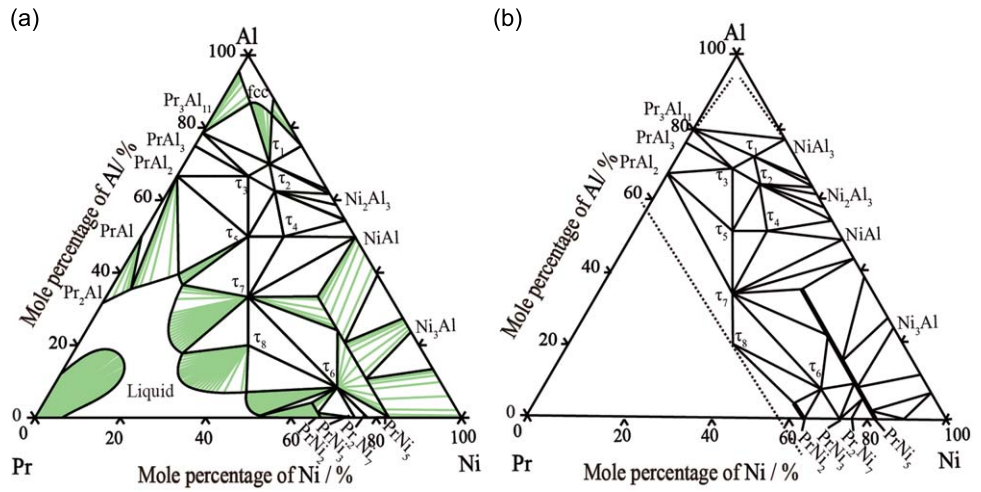

Figure 3. Calculated isothermal section of the Ni–Al–Pr ternary system at 800 °C compared with the experimental results. (a) Calculated, (b) experimental [35].

Table 2. Optimized thermodynamic parameters in this work.

Phase		Thermodynamic parameters
	Liquid	${}^0L^{\mathrm{Liq}}_{\mathrm{Al,Ni,Pr}} = -273268,\ {}^1L^{\mathrm{Liq}}_{\mathrm{Al,Ni,Pr}} = 0,\ {}^2L^{\mathrm{Liq}}_{\mathrm{Al,Ni,Pr}} = 0\ {}^0L^{\mathrm{Liq}}_{\mathrm{Al,Ni:Pr}} = -306268$
	fcc	${}^0L^{\mathrm{fcc}}_{\mathrm{Al,Pr}} = -470000,\ {}^1L^{\mathrm{fcc}}_{\mathrm{Al,Pr}} = -50000$
	dhcp	$L^{\mathrm{dhcp}}_{\mathrm{Al,Pr}} = -93322 + 40.5T$
	$\alpha^{\mathrm{Al}}_{\mathrm{Al_{11}Pr_3}}$	${}^0G^{\alpha-\mathrm{Al_{11}Pr_3}}_{\mathrm{Pr:Al}} = 3{}^0G^{\mathrm{dhcp}}_{\mathrm{Pr}} + 11\ {}^0G^{\mathrm{fcc}}_{\mathrm{Al}} - 592350 + 120.29T$
	$\beta^{\mathrm{Al}}_{\mathrm{Al_{11}Pr_3}}$	${}^0G^{\beta-\mathrm{Al_{11}Pr_3}}_{\mathrm{Pr:Al}} = 3{}^0G^{\mathrm{dhcp}}_{\mathrm{Pr}} + 11\ {}^0G^{\mathrm{fcc}}_{\mathrm{Al}} - 593588 + 119.29T$
	Ni_5Pr	${}^0G^{\mathrm{Ni_5Pr}}_{\mathrm{Al:Pr}} = {}^0G^{\mathrm{dhcp}}_{\mathrm{Pr}} + 5\ {}^0G^{\mathrm{fcc}}_{\mathrm{Ni}} - 24000,\ L^{\mathrm{Ni_5Pr}}_{\mathrm{Al,Ni:Pr}} = -166000 + 30T$
	$\mathrm{Ni_{2Pr}}$	${}^0G^{\mathrm{Ni_2Pr}}_{\mathrm{Al:Pr}} = {}^0G^{\mathrm{dhcp}}_{\mathrm{Pr}} + 2\ {}^0G^{\mathrm{fcc}}_{\mathrm{Ni}} + 100000,\ L^{\mathrm{Ni_2Pr}}_{\mathrm{Al,Ni:Pr}} = -273259 + 15T$
τ_1	Ni_2Al_7Pr	${}^0G^{\mathrm{Ni_2Al_7Pr}}_{\mathrm{Al:Ni:Pr}} = 7{}^0G^{\mathrm{fcc}}_{\mathrm{Al}} + {}^0G^{\mathrm{dhcp}}_{\mathrm{Pr}} + 2{}^0G^{\mathrm{fcc}}_{\mathrm{Ni}} - 430041 - 18T$
τ_2	Ni_2Al_5Pr	${}^0G^{\mathrm{Ni_2Al_5Pr}}_{\mathrm{Al:Ni:Pr}} = 5{}^0G^{\mathrm{fcc}}_{\mathrm{Al}} + {}^0G^{\mathrm{dhcp}}_{\mathrm{Pr}} + 2{}^0G^{\mathrm{fcc}}_{\mathrm{Ni}} - 420033 - 14.93T$
τ_3	$NiAl_4Pr$	${}^0G^{\mathrm{Ni\ Al_4Pr}}_{\mathrm{Al:Ni:Pr}} = 4{}^0G^{\mathrm{fcc}}_{\mathrm{Al}} + {}^0G^{\mathrm{dhcp}}_{\mathrm{Pr}} + {}^0G^{\mathrm{fcc}}_{\mathrm{Ni}} - 299482 + 3T$
τ_4	Ni_2Al_3Pr	${}^0G^{\mathrm{Ni_2Al_3Pr}}_{\mathrm{Al:Ni:Pr}} = 3{}^0G^{\mathrm{fcc}}_{\mathrm{Al}} + {}^0G^{\mathrm{dhcp}}_{\mathrm{Pr}} + 2{}^0G^{\mathrm{fcc}}_{\mathrm{Ni}} - 374500 + 2.12T$
τ_5	$NiAl_2Pr$	${}^0G^{\mathrm{Ni\ Al_2Pr}}_{\mathrm{Al:Ni:Pr}} = 2{}^0G^{\mathrm{fcc}}_{\mathrm{Al}} + {}^0G^{\mathrm{dhcp}}_{\mathrm{Pr}} + {}^0G^{\mathrm{fcc}}_{\mathrm{Ni}} - 264850 + 10.33T$
τ_6	Ni_8AlPr_3	${}^0G^{\mathrm{Ni_8Al\ Pr_3}}_{\mathrm{Al:Ni:Pr}} = {}^0G^{\mathrm{fcc}}_{\mathrm{Al}} + 3{}^0G^{\mathrm{dhcp}}_{\mathrm{Pr}} + 8{}^0G^{\mathrm{fcc}}_{\mathrm{Ni}} - 463090 - 40T$
τ_7	NiAlPr	${}^0G^{\mathrm{Ni\ Al\ Pr}}_{\mathrm{Al:Ni:Pr}} = {}^0G^{\mathrm{fcc}}_{\mathrm{Al}} + {}^0G^{\mathrm{dhcp}}_{\mathrm{Pr}} + {}^0G^{\mathrm{fcc}}_{\mathrm{Ni}} - 186291 - 12.1T$
τ_8	Ni_2AlPr_2	${}^0G^{\mathrm{Ni_2Al\ Pr_2}}_{\mathrm{Al:Ni:Pr}} = 2{}^0G^{\mathrm{fcc}}_{\mathrm{Al}} + {}^0G^{\mathrm{dhcp}}_{\mathrm{Pr}} + 2{}^0G^{\mathrm{fcc}}_{\mathrm{Ni}} - 280085 + 0.7T$

$$ {}^{\mathrm{ex}}G^{\varphi}_m = x_{\mathrm{Al}}x_{\mathrm{Ni}}L^{\varphi}_{\mathrm{Al,Ni}} + x_{\mathrm{Al}}x_{\mathrm{Pr}}L^{\varphi}_{\mathrm{Al,Pr}} + x_{\mathrm{Pr}}x_{\mathrm{Ni}}L^{\varphi}_{\mathrm{Pr,Ni}} + x_{\mathrm{Al}}x_{\mathrm{Ni}}x_{\mathrm{Pr}}L^{\varphi}_{\mathrm{Al,Ni,Pr}} \tag{2} $$

where $L^{\varphi}_{\mathrm{Al,Ni}}$, $L^{\varphi}_{\mathrm{Al,Pr}}$ and $L^{\varphi}_{\mathrm{Pr,Ni}}$ are the binary interaction parameters and $L^{\varphi}_{\mathrm{Al,Ni,Pr}}$ corresponds to the Ni–Al–Pr ternary interaction parameters that are evaluated in this work.

3.2. Binary compounds

Thermodynamic models for Ni_5Pr and Ni_2Pr need to be updated because of the solubility of Al in this phase and are expressed as:

$$ G^{(Al,Ni)_m Pr_n} = y_{Al}G_{Al:Pr} + y_{Ni}G_{Ni:Pr} + mRT(y_{Al}\ln(y_{Al}) + y_{Ni}\ln(y_{Ni})) + y_{Al}y_{Ni}L_{\mathrm{Pr:Al,Ni}} \tag{3} $$

in which y_{Al} and y_{Ni} are the site fractions of Al and Ni in the first sublattice, m and n are stoichiometric numbers, $G_{\mathrm{Al:Pr}}$ and $G_{\mathrm{Pr:Ni}}$ are the Gibbs free energies for end-members and $L_{\mathrm{Pr:Al,Ni}}$ is the interaction term between the two sublattices to be optimized.

3.3. Ternary compounds

Eight ternary intermetallic compounds, Ni_2Al_7Pr, Ni_2Al_5Pr, $NiAl_4Pr$, Ni_2Al_3Pr, $NiAl_2Pr$, Ni_8AlPr_3, NiAlPr and Ni_2AlPr_2 were treated as stoichiometric compounds because no available experimental data were reported on the homogeneity range for this ternary phase. The Gibbs free energy of $Ni_mAl_nPr_l$ is:

Table 3. Calculated invariant reactions of the Al–Pr binary system.

Invariant reaction	Temperature, °C			at % Pr in liquid	
	This work	Yin's work [28]	Experiment	This work	Experiment
Liq → fcc + α_$Al_{11}Pr_3$	639	639	640 [19]	2.054	5 [19]
Liq + Al_2Pr → β_$Al_{11}Pr_3$	1246	1245	1240 [19]	18.94	20.5 [19]
β_$Al_{11}Pr_3$ + Al_2Pr → Al_3Pr	1075		1075 [19]	33.33	33.3 [19]
Liq + Al_2Pr → AlPr	899	899	905 [19]	64.41	66.0 [19]
β_$Al_{11}Pr_3$ → Al_3Pr + α_$Al_{11}Pr_3$	965		965 [19]	21.42	21.42 [19]
Liq + β_$Al_{11}Pr_3$ → α_$Al_{11}Pr_3$	965		965 [19]	7.50	21.42 [19]
Liq + AlPr → $AlPr_2$	739	735	739 [19, 22]	70.41	67.8 [19] , 70 [22]
Liq + $AlPr_2$ → $AlPr_3$	684	684	685 [22]	76.02	76.0 [22]
Liq → $AlPr_3$ + bcc	651	651	650 [22]	81.41	82.5 [22]
bcc → $AlPr_3$ + dhcp	622	619	620 [22]	91.45	92.0 [22]

Table 4. Calculated enthalpies of formation of ternary intermetallic compounds .

No.	Phase	Enthalpy of formation/kJ mol^{-1}				
		Measured(300 K)	Miedema model	OQMD[52]	Materials project[53]	Present work
τ_1	Ni_2Al_7Pr	…	−40.61	…	…	−40.91
τ_2	Ni_2Al_5Pr	…	−49.56	−57.02	−56.44	−52.50
τ_3	$NiAl_4Pr$	…	−49.71	−52.02	−51.83	−49.91
τ_4	Ni_2Al_3Pr	…	−61.34	−58.17	−57.21	−62.41
τ_5	$NiAl_2Pr$	…	−66.32	…	−55.02	−66.21
τ_6	Ni_8AlPr_3	…	−37.18	…	−30.96	−38.59
τ_7	NiAlPr	−56 ± 3 [38]	−66.97	−47.60	−46.54	−62.07
τ_8	Ni_2AlPr_2	…	−54.19	−40.48	−38.37	−56.01

$$G^{Ni_mAl_nPr_l}_{Al:Ni:Pr} = nG^{fcc}_{Al} + lG^{dhcp}_{Pr} + mG^{fcc}_{Ni} + A + BT \tag{4}$$

where G^{fcc}_{Al}, G^{fcc}_{Ni} and G^{dhcp}_{Pr} are the molar Gibbs free energy for pure elements Al, Ni and Pr in the structure of fcc, fcc and dhcp. Thermodynamic parameters A and B are evaluated in this work. Ouyang's model with Miedema theory was used to calculate the formation enthalpy of ternary intermetallic compounds, and it can be expressed as:

$$\Delta H_{ABC} = \frac{C_A C_B}{X_{AB}X_{BA}}\Delta H_{AB}(X_{AB}, X_{BA}) + \frac{C_A C_C}{X_{AC}X_{CA}}\Delta H_{AC}(X_{AC}, X_{CA}) + \frac{C_B C_C}{X_{BC}X_{CB}}\Delta H_{BC}(X_{BC}, X_{CB}) \tag{5}$$

$$\begin{aligned} X_{AB} &= C_A + C_C\frac{\lambda_A}{\lambda_A+\lambda_B} \quad X_{BA} = C_B + C_C\frac{\lambda_B}{\lambda_A+\lambda_B} \\ X_{AC} &= C_A + C_B\frac{\lambda_A}{\lambda_A+\lambda_C} \quad X_{CA} = C_C + C_B\frac{\lambda_C}{\lambda_A+\lambda_C} \\ X_{BC} &= C_B + C_A\frac{\lambda_B}{\lambda_B+\lambda_C} \quad X_{CB} = C_C + C_A\frac{\lambda_C}{\lambda_B+\lambda_C} \\ \lambda_A &= [\Delta H^{inter}_{C\ in\ A} - \Delta H^{inter}_{B\ in\ A}]^2 \quad \lambda_B = [\Delta H^{inter}_{A\ in\ B} - \Delta H^{inter}_{C\ in\ B}]^2 \\ \lambda_C &= [\Delta H^{inter}_{B\ in\ C} - \Delta H^{inter}_{A\ in\ C}]^2 \end{aligned} \tag{6}$$

in which ΔH_{ABC} is the formation enthalpy of the ternary phase. ΔH_{AB}, ΔH_{AC} and ΔH_{BC} are the formation enthalpies of the binary phases. C_A, C_B and C_C are mole fractions of the components in the ternary system with $C_A + C_B + C_C = 1$, and $\Delta H^{inter}_{A\ in\ B}$ is the enthalpy of component A dissolved in B, which is obtained by Miedema theory and is expressed by:

$$\Delta H^{inter}_{B\ in\ A} = \frac{V_B^{\frac{2}{3}}}{(n_{WS}^{-\frac{1}{3}})_{av}}\left\{-P(\Delta\varnothing)^2 + Q(\Delta n_{WS}^{\frac{1}{3}})^2 - R)\right\} \tag{7}$$

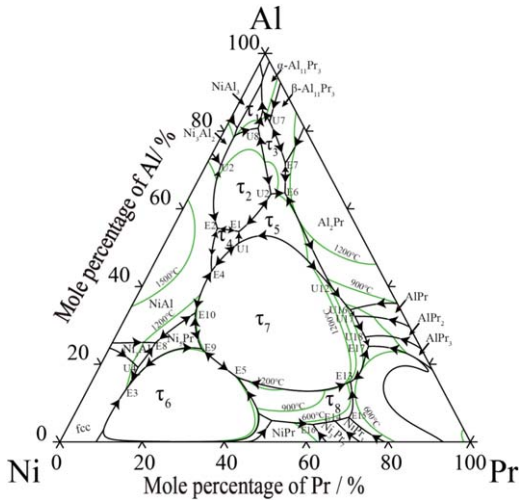

Figure 4. Calculated liquidus projection of the Ni–Al–Pr ternary system.

$$(n_{WS}^{-\frac{1}{3}})_{av} = \frac{1}{2}\left\{\left(\frac{1}{(n_{WS}^{A})^{\frac{1}{3}} + (n_{WS}^{B})^{\frac{1}{3}}}\right)\right\} \tag{8}$$

where V, n_{WS} and $\Delta\varnothing$ represent the molar volume, electron-density discontinuity and electronegativity difference. P, Q, a and R are empirical constants [23], and their values are obtained from [57].

4. Results and discussion

4.1. Binary system

Calculated Ni–Al and Ni–Pr phase diagrams based on Huang's [33] and Rahou's [32] thermodynamic parameters are shown in figures 1(a) and (b), respectively.

Figure 2 shows the calculated equilibrium phase diagram of the Al–Pr binary system with experimental data. Optimized parameters for the Al–Pr system based on Yin's [28] thermodynamic parameters are presented in table 2. We have re-optimized the parameter of the dhcp solution phase because Yin's optimization parameter of the dhcp solution phase cannot reproduce the equilibrium that is related to the dhcp phase. For the $Al_{11}Pr_3$ phase, the allotropic transformation was included, and the transformation temperature between the α-$Al_{11}Pr_3$ and β-$Al_{11}Pr_3$ phases is 965 °C, which is the same as that in the literature [19]. The temperatures of the related invariant reactions, Liq + Al_2Pr → β-$Al_{11}Pr_3$ and β-$Al_{11}Pr_3$ + Al_2Pr → Al_3Pr were 1246 °C and 1075 °C, respectively, which is consistent with the experimental data. The calculated solubility of Pr in the fcc (Al) phase is considered to be ~0.008 at% at 620 °C, which is consistent with the experimental data [21] in figure 2(b). The temperature of the eutectic reaction Liq → Fcc + α-$Al_{11}Pr_3$ remained changed. All invariant reactions are presented in table 3 in contrast with the literature. The deviation between calculated and experimental results was less than 5 °C. In general, our calculated phase-equilibrium result agreed well with the experimental data.

4.2. Ternary system

Table 2 gives the optimized thermodynamic parameters for the Ni–Al–Pr ternary system. The enthalpy of formation of the ternary compounds has been calculated using the CALPHAD method and Ouyang's models with Miedema theory, and these are compared with different sources in table 4. For the τ_1, τ_2, τ_3, τ_4 phases, the calculated results are within 5 kJ mol^{-1} deviations from both the values of Miedema model and DFT method. Taking the τ_5, τ_6, τ_8 phases for example, the calculated enthalpies of formation that were obtained using the CALPHAD method were matched with the values that were calculated using the Miedema model, and some deviation from the DFT calculation results of OQMD [52] and Materials Project [53] database. Although we have tried hard to adjust the parameters to make the formation enthalpy of the τ_7 phase consistent with the experimental values [38], it still remains some deviation from the experimental measurement.

The phase-equilibrium information at 800 °C that was calculated using these optimized parameters is shown in figure 3(a) and is compared with the experimental data in figure 3(b) [35]. The calculated solubilities of Al in Ni_5Pr and Ni_2Pr were 33.5 at% and 7.3 at%, respectively, which consistent with the experimental data. Liquid phase has been predicted at Al-Rich corner although it has not been observed in experiments. According to the

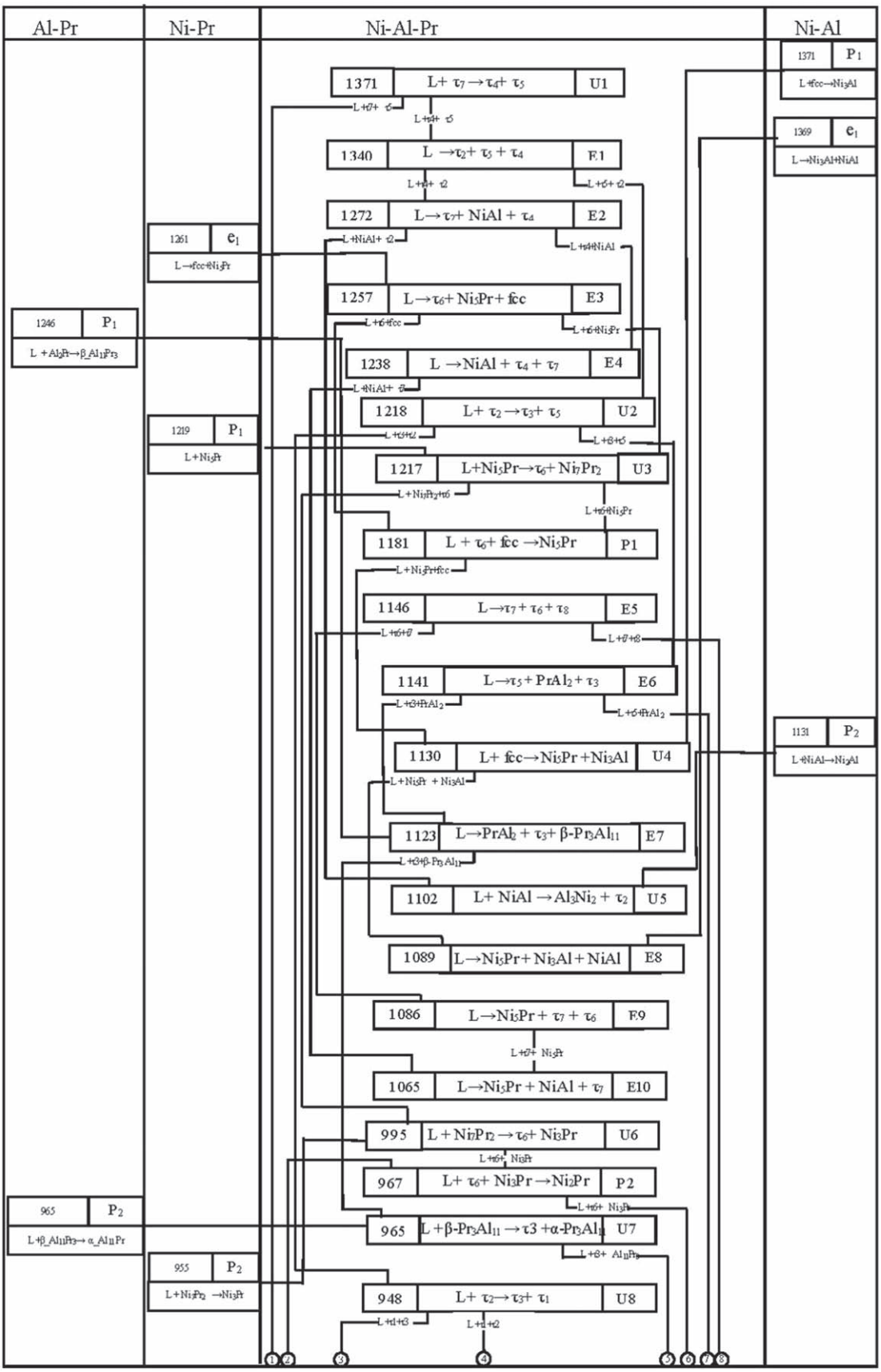

Figure 5. Reaction scheme of Ni–Al–Pr ternary system. Temperature is in °C.

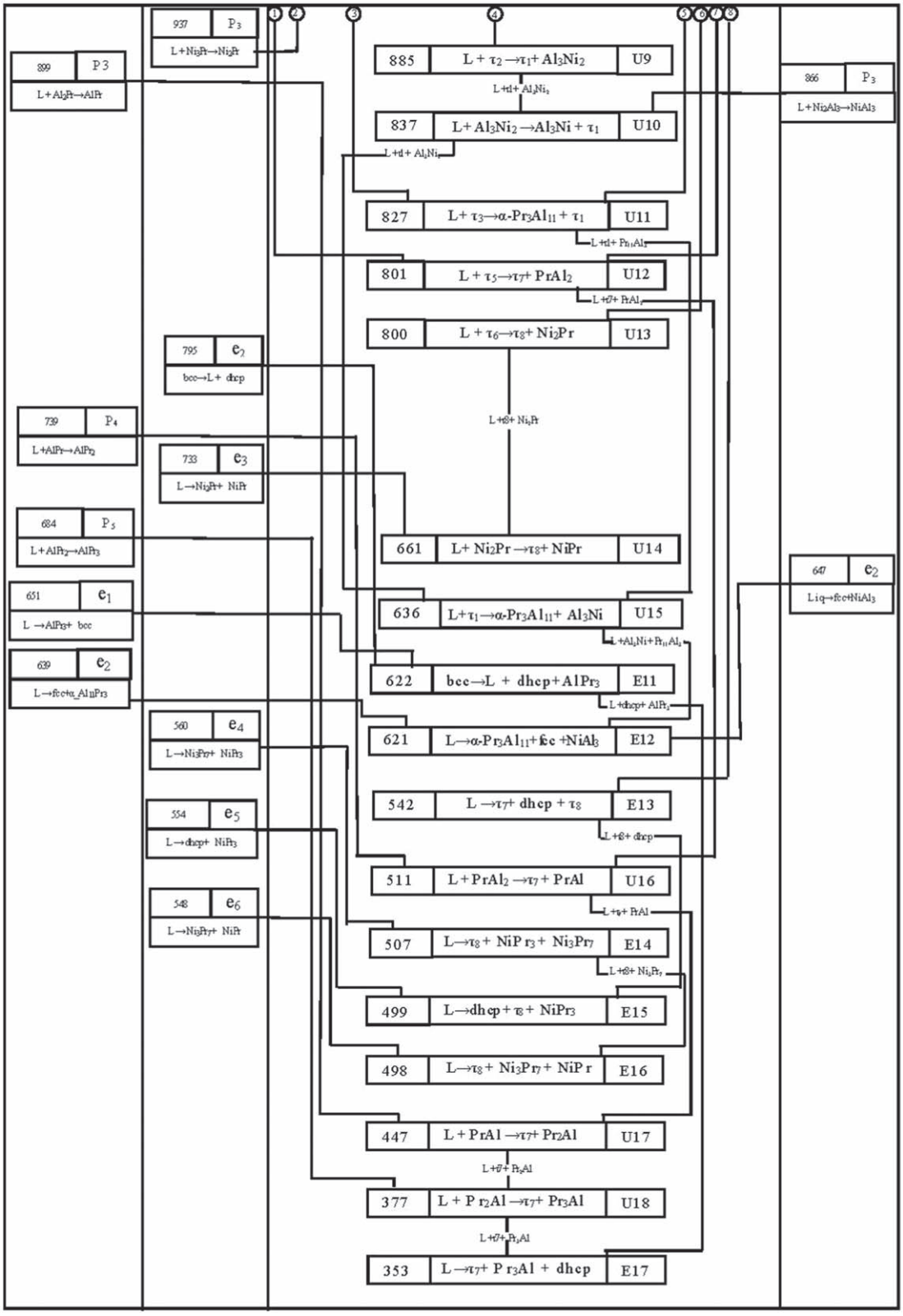

Figure 5. (Continued.)

Ni–Al–Nd ternary system measured by Gödecke *et al* [58] at 700 °C using metallography analysis, thermal and magnetothermal analysis and x-ray analysis, the liquid phase was observed at the Al-rich corner. Our previous works [15, 16] calculated the isothermal section at 800 °C of Ni–Al–Y and Ni–Al–Ce ternary systems, it has found that the liquid phase was also predicted at Al-Rich corner. From this we can see that the results of our calculations are reasonable. In the Pr-Rich corner, liquid phase is predicted to be presence. This is probably due to lack of ternary compound in this region and deserves further determination in the future. It should be noted

Table 5. Calculated invariant reaction of the Ni–Al–Pr ternary system.

No.	Invariant reaction	Temperature, °C	Composition, %		
			Al	Ni	Pr
U1	Liq + $\tau_7 \rightarrow \tau_4 + \tau_5$	1371	51.2	30.9	18.0
E1	Liq $\rightarrow \tau_2 + \tau_5 + \tau_4$	1340	54.2	29.3	16.5
E2	Liq $\rightarrow \tau_2$ + NiAl + τ_4	1272	55.0	34.1	10.9
E3	Liq $\rightarrow \tau_6$ + Ni_5Pr + fcc	1257	3.1	87.8	9.1
E4	Liq $\rightarrow$ NiAl + $\tau_4 + \tau_7$	1238	44.0	41.1	15.0
U2	Liq + $\tau_2 \rightarrow \tau_3 + \tau_5$	1218	63.8	16.7	19.5
U3	Liq + $Ni_5Pr \rightarrow \tau_6 + Ni_7Pr_2$	1217	0.1	72.2	27.7
P1	Liq + τ_6 + fcc $\rightarrow Ni_5Pr$	1181	14.5	76.0	9.5
E5	Liq $\rightarrow \tau_7 + \tau_6 + \tau_8$	1146	16.5	48.4	35.1
E6	Liq $\rightarrow \tau_5 + PrAl_2 + \tau_3$	1141	64.1	13.1	22.8
U4	Liq + fcc$\rightarrow Ni_5Pr + Ni_3Al$	1130	19.2	71.4	9.4
E7	Liq $\rightarrow PrAl_2 + \tau_3 + \beta_Pr_3Al_{11}$	1123	71.8	9.2	19.0
U5	Liq + NiAl $\rightarrow Al_3Ni_2 + \tau_2$	1102	71.2	25.2	3.6
E8	Liq $\rightarrow Ni_5Pr + Ni_3Al$ + NiAl	1089	25.5	63.9	10.6
E9	Liq $\rightarrow Ni_5Pr + \tau_7 + \tau_6$	1086	23.9	53.5	22.6
E10	Liq $\rightarrow Ni_5Pr$ + NiAl + τ_7	1065	33.2	50.4	16.4
U6	Liq + $Ni_7Pr_2 \rightarrow \tau_6 + Ni_3Pr$	995	0.0	64.5	35.4
P2	Liq + $\tau_6 + Ni_3Pr \rightarrow Ni_2Pr$	967	0.1	63.7	36.2
U7	Liq $+\beta_Pr_3Al_{11} \rightarrow \tau_3 + \alpha_Pr_3Al_{11}$	965	82.4	7.7	9.9
U8	Liq + $\tau_2 \rightarrow \tau_3 + \tau_1$	948	80.7	11.3	8.0
U9	Liq + $\tau_2 \rightarrow \tau_1 + Al_3Ni_2$	885	78.8	18.0	3.2
U10	Liq + $Al_3Ni_2 \rightarrow Al_3Ni + \tau_1$	837	80.8	16.4	2.8
U11	Liq + $\tau_3 \rightarrow \alpha_Pr_3Al_{11} + \tau_1$	827	85.2	8.0	6.8
U12	Liq + $\tau_5 \rightarrow \tau_7 + PrAl_2$	801	39.6	13.8	46.6
U13	Liq + $\tau_6 \rightarrow \tau_8 + Ni_2Pr$	800	7.3	47.9	44.8
U14	Liq + $Ni_2Pr \rightarrow \tau_8$ + NiPr	661	5.5	45.7	48.8
U15	Liq + $\tau_1 \rightarrow \alpha_Pr_3Al_{11} + Al_3Ni$	636	92.3	4.7	3.0
E11	Liq $\rightarrow \alpha_Pr_3Al_{11}$ + fcc + τ_1	621	93.8	3.6	2.6
E12	bcc $\rightarrow$ Liq+dhcp + Pr_3Al	622	21.3	2.3	76.4
E13	Liq $\rightarrow \tau_7$ + dhcp + τ_8	542	15.5	21.4	63.1
U16	Liq + $PrAl_2 \rightarrow \tau_7$ + PrAl	511	34.0	12.3	53.7
E14	Liq $\rightarrow \tau_8 + NiPr_3 + Ni_3Pr_7$	507	5.4	30.3	64.3
E15	Liq $\rightarrow$ dhcp + $\tau_8 + NiPr_3$	499	6.3	26.4	67.2
E16	Liq $\rightarrow \tau_8 + Ni_3Pr_7$ + NiPr	498	4.5	36.1	59.4
U17	Liq + PrAl $\rightarrow \tau_7 + Pr_2Al$	447	32.3	12.0	55.8
U18	Liq + $Pr_2Al \rightarrow \tau_7 + Pr_3Al$	377	27.0	12.1	60.9
E17	Liq $\rightarrow \tau_7 + Pr_3Al$ + dhcp	353	24.6	12.5	62.9

that the calculated phase-equilibrium at the nickel-Rich corner differ from the experimental data. The calculated phase equilibria compounds were Ni_5Pr and fcc(Ni) rather than Ni_5Pr, fcc(Ni) and $Ni_{17}Pr_2$. This is because the $Ni_{17}Pr_2$ binary compound cannot exist in the Ni–Pr binary system as studied by Huang *et al* [29] and Pan *et al* [30], and the Rahou's [32] such that thermodynamic description contains no information of the $Ni_{17}Pr_2$ binary compound. Overall, our calculated phase equilibria are in good agreement the measured data.

A liquidus projection has been calculated with isothermal lines in figure 4 and the Scheil reaction scheme for the Ni–Al–Pr ternary system is illustrated in figure 5. Because of too many reactions, some phases in figure 5 have been expressed in simplified way to simplify the reading of the scheme. Details of composition and temperature of calculated invariant reactions for the Ni–Al–Pr ternary system are given in table 5. A eutectic reaction Liq $\rightarrow \tau_1$ + fcc + α-$Al_{11}Pr_3$ existed in the Al-rich corner, and the calculated temperature was 621 °C. Gödecke *et al* determined a similar eutectic reaction for the Ni–Al–La [58] and Ni–Al–Nd ternary system at the Al-rich corner [59] by using metallography analysis, thermal and magnetothermal analysis and x-ray analysis. The measured temperatures of the Liq $\rightarrow \tau_1$ + fcc + α-$Al_{11}La_3$ and Liq $\rightarrow \tau_1$ + fcc + $\alpha_Al_{11}Nd_3$ were 625 °C and 626 °C, respectively. From these results, we confirm that our calculated results are reasonable.

5. Conclusions

Based on the experimental information, the phase relationships and thermodynamics of the Ni–Al–Pr ternary system have been assessed using the CALPHAD approach. Eight ternary intermetallic compounds were

considered and the formation enthalpy of the ternary phase was predicted by using Ouyang's model with Miedema theory. The solubilities of the Ni_2Pr and Ni_5Pr have also been considered. Calculated phase-equilibria information of the isothermal condition at 800°C is consistent with the experimental data. The re-optimized Al–Pr binary system, which considered the solubility of Pr in fcc (Al), is fitted fairly well by the experimental data. A self-consistent set of thermodynamic parameters of the Ni–Al–Pr and Al–Pr system were derived. As the rare earth elements always co-exist in nature which is hard to separate from each other., the present work provides more information for further potential application of Pr (together with other REEs) into the Ni–Al-alloys.

Acknowledgments

This work is financially supported by National Natural Science Foundation of China (Grant No. 51761014, U1537212, 51871248), Jiangxi Provincial Department of Science and Technology (Grant No. 20171BAB216004), The Youth Jinggang Scholars Program in Jiangxi Province, and Ganzhou Talents Program.

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